Glossary and Acronyms

Absorption: The assimilation of one substance by another.

Accuracy: Often confused with precision. Refers to the agreement of a reading or observation obtained from an instrument with the true value.

Action Level: Term used by OSHA and NIOSH to express the exposure level of a chemical which requires medical surveillance, usually one-half the PEL or REL.

Activated Charcoal: Charcoal is an amorphous form of carbon formed by burning wood, nutshells, animal bones, and other carbonaceous materials. Charcoal becomes activated by heating it with steam to 800–900°C. During this treatment, a porous, submicroscopic internal structure is formed which gives it an extensive internal surface area. Activated charcoal is commonly used as a gas or vapor adsorbant in air-purifying respirators and as a solid sorbant in air-sampling.

Activation: Making a substance artificially radioactive in an accelerator or by bombarding it with protons or neutrons in a reactor.

Administrative Controls: Methods of controlling employee exposures by job rotation, work assignment, or time periods away from the hazard.

Adsorption: The condensation of gases, liquids, or dissolved substances on the surfaces of solids.

Aerodynamic Diameter: The diameter of a unit density sphere having the same settling velocity as the particle in question of whatever shape and density.

Aerosols: Liquid droplets or solid particles dispersed in air, that are of fine enough particle size (0.01 to 100 micrometers) to remain so dispersed for a period of time.

Air-Purifying Respirator: Respirators that use filters or sorbents to remove harmful substances from the air.

Air-Supplied Respirator: Respirator that provides a supply of breathable air from a clean source outside of the contaminated work area.

Alignment: The positioning of a lithography mask in relation to a wafer.

Alpha Particle: One type of particle which can be emitted by a radioactive nucleus. It is similar to the nucleus of a helium atom and consists of two protons and two neutrons.

Alveoli: Tiny air sacs of the lungs, formed at the ends of bronchioles; through the thin walls of the alveoli, the blood takes in oxygen and gives up its carbon dioxide in the process of respiration.

American Conference of Governmental Industrial Hygienists (ACGIH): A professional society of governmental and academic industrial hygienists which is devoted to the development of administrative and technical aspects of worker health protection.

Amorphous: Noncrystalline.

Anaerobic Bacteria: Any bacteria that can survive in partial or complete absence of air.

Anisotropic Etching: refers to any etching where the material is removed only in a downward direction—as opposed to isotropic etching. Often associated with dry etch processes.

ANSI: American National Standards Institute; a voluntary membership organization (run with private funding) that develops consensus standards nationally for a wide variety of devices and procedures.

Article: Article means a manufactured item: *(i)* which is formed to a specific shape or design during manufacture; *(ii)* which has end use function(s) dependent in whole or in part upon its shape or design during end use; and *(iii)* which does not release, or otherwise result in exposure to, a hazardous chemical, under normal conditions of use.

ASHRAE: American Society of Heating, Refrigerating and Air-conditioning Engineers; a professional organization that generates voluntary consensus standards on heating, refrigeration and air-conditioning.

Assigned Protection Factor: The expected level of respiratory protection that would be provided by a properly functioning respirator or class of respirators, to a stated percentage of properly fitted and trained users.

Audible Range: The frequency range over which normal ears hear—approximately 20 Hz through 20,000 Hz. Above the range of 20,000 Hz, the term ultrasonic is used. Below 20 Hz, the term subsonic is used.

Audible Sound: Sound containing frequency components lying between 20 and 20,000 Hz.

Audiogram: A record of hearing loss of hearing level measured at several different frequencies—usually 500 to 6000 Hz. The audiogram may be presented graphically or numerically. Hearing level is shown as a function of frequency.

Balancing By Dampers: Method for designing local exhaust system ducts using adjustable dampers to distribute airflow after installation.

Beam Divergence: Angle of beam spread measured in milliradians (1 milliradian = 4 minutes of arc).

Becquerel: A unit of radioactivity. One becquerel is the activity corresponding to one disintegration per second.

Benign: Not malignant. A benign tumor is one which does not metastasize or invade tissue. Benign tumors may still be lethal, due to pressure on vital organs.

Berylliosis: Chronic beryllium intoxication.

Beta Particle: One type of particle which can be emitted by a radioactive nucleus. It is similar to an orbital electron and can be negatively or positively charged.

Bioaerosols: Airborne particles, large molecules, or volatile compounds that are living or were released from living organisms.

Biohazard: A combination of the words biological hazard. Organisms or products of organisms that present a risk to humans.

Biological Agent: Includes infectious microorganisms, toxic biological substances, and biological allergens.

Biological Half-life: The time required to reduce the amount of an exogenous substance in the body by half.

Biological Monitoring: A measurement of exposure taken from biological media (body fluid or gas), usually of the concentration of a chemical/metabolite or the status of a bio-marker; normally collected with the intent of relating the measured value to the absorbed dose of a chemical over some averaging period.

Boat: A holder for wafers used during different semiconductor manufacturing tasks. The holder is made of Teflon or plastic when used in wet chemical processes. It is made of quartz when used in high-temperature operations.

Breathing Zone Sample: An air-sample collected in the breathing zone of workers to assess their exposure to airborne contaminants.

Breathing Zone: Imaginary globe of two foot radius surrounding the head.

Bremsstrahlung: Secondary x-radiation produced when a beta particle is slowed down or stopped by a high-density surface.

Bubbler: A liquid-containing apparatus used to supply vapors to a process tool such as a diffusion furnace by bubbling a carrier gas through the liquid.

Buffer: Any substance in a fluid which tends to resist the change in pH when acid or alkali is added.

Buffered Oxide Etch: A wet chemical mixture of hydrogen fluoride (HF) and ammonium fluoride (NH_4F) used to etch an oxide layer.

Capture Velocity: Air velocity at any point in front of the hood necessary to overcome opposing air currents and to capture the contaminated air by causing it to flow into the exhaust hood.

Carpal Tunnel: A passage in the wrist through which the median nerve and many tendons pass to the hand from the forearm.

Carpal Tunnel Syndrome: A common affliction caused by compression of the median nerve in the carpal tunnel. Often associated with tingling, pain, or numbness in the thumb and first three fingers.

Carrier Gas: A gas, inert with respect to the specific manufacturing process, used to transport another gas or vapor to a reaction chamber.

CAS Number: Identifies a particular chemical by the Chemical Abstract Service, a service of the American Chemical Society that indexes and compiles abstracts of worldwide chemical literature called *Chemical Abstracts*.

Ceiling Limit (C): An airborne concentration of a toxic substance in the work environment, which should never be exceeded.

Chemical Cartridge: The type of absorption unit used with a respirator for removal of low concentrations of specific vapors and gases.

Chemical Etching: Same as wet etching.

Chip: An individual die or semiconductor device.

Class Number: (1) Number of particles above a specific size in a cubic foot of air in the clean room. *(2)* The hazard classification of a laser on a scale of 1 to 4 with 1 being the lowest hazard classification and 4 the highest.

Cleanroom: An area within a semiconductor manufacturing facility where air and surface contaminates are tightly controlled for process reasons.

CO: Carbon monoxide; a product of incomplete combustion.

CO_2: Carbon dioxide; produced by an individuals' respiration; a concentration greater than 800–1000 ppm is generally an indicator of inadequate ventilation.

Combustible Liquids: Combustible liquids are those having a flash point at or above 100˚F (37.8˚C).

Contact Dermatitis: Dermatitis caused by contact with a substance—gaseous, liquid, or solid. May be due to primary irritation or an allergy.

Cryogenic Pump: A vacuum pump that produces a vacuum down to 10^{-10} torr.

Cumulative Trauma Disorder (CTD): CTDs are a class of musculoskeletal disorders involving damage to the tendons, tendon sheaths, synovial lubrication of the tendon sheaths, and the related bones, muscles, and nerves of the hands, wrists, elbows, shoulders, neck, and back. The more frequently occurring occupationally-induced disorders in this class include carpal tunnel syndrome, epicondylitis (tennis elbow), tendonitis, synovitis, stenosing tenosynovitis of the finger, DeQuervain's Disease, and low back pain.

Curie: A unit of radioactivity. One curie is the activity corresponding to a disintegration rate of 3.7×10^{10} disintegrations per second.

Cutie-Pie: A portable instrument equipped with a direct-reading meter used to determine the level of ionizing radiation in an area. Has a fairly flat response at different energy levels making it good as a direct-reading instrument.

CVD: Chemical vapor deposition; a semiconductor manufacturing process step where chemical vapors are used to form a new conductor, dielectric, or semiconductor layer on a wafer.

CW Laser: A continuous-wave laser. A laser whose output lasts for a comparatively long uninterrupted time period, while the laser is actuated. Occasionally meant to mean a laser emitting for a period in excess of 0.25 seconds.

Dampers: Adjustable sources of airflow resistance used to distribute airflow in a ventilation system.

dBA: Sound level in decibels read on the A-scale of a sound-level meter. The A scale discriminates against very low frequencies (as does the human ear) and is, therefore, better for measuring general sound levels. See also *Decibel*.

dBC: Sound level in decibels read on the C-scale of a sound-level meter. The C scale discriminates very little against very low frequencies. See also *Decibel*.

Decay: When a radioactive atom disintegrates, it is said to decay. What remains is a different element. An atom of polonium decays to form lead, ejecting an alpha particle in the process.

Decibel (dB): A unit used to express sound power level (L_w). Sound power is the total acoustic output of a sound source in watts (W). By definition, sound power level, in decibels, is: L_w = 10 log W divided by W_o were W is the sound power of the source and W_o is the reference sound power.

Dermatitis: Inflammation of the skin from any cause.

Dermatosis: A broader term than dermatitis; it includes any cutaneous abnormality, thus encompassing folliculitis, acne, pigmentary changes, and nodules and tumors.

Developer: A chemical used to remove areas defined in the masking and exposure steps of wafer fabrication.

DI Water: Deionized water used in semiconductor processing because of its purity.

Diffusion: A high temperature semiconductor manufacturing process where molecules of certain chemicals are introduced into a substrate such as silicon.

Direct-Reading Instrumentation: Those instruments that give an immediate indication of the concentration of aerosols, gases, or vapors or magnitude of physical hazard by some means such as a dial or meter.

Discrete Device: A single-component circuit such as a diode, transistor, etc. as opposed to an integrated circuit.

Disinfectant: An agent that frees from infection by killing the vegetative cells of microorganisms.

Disintegration: A nuclear transformation or decay process that results in the release of energy in the form of radiation.

Dispersion: The general term describing systems consisting of particulate matter suspended in air or other fluid; also, the mixing and dilution of contaminant in the ambient environment.

Dopant: An element (e.g., arsenic, boron, phosphorous...) used to change the electrical conductivity of a semiconductor material.

Doping: Introducing dopant elements into the crystal lattice of a semiconductor through either a diffusion process or through ion implantation.

Dose: (1) A term used to express the amount of a chemical or of ionizing radiation energy absorbed in a unit volume or an organ or individual. Dose rate is the dose delivered per unit of time. (See also *Roentgen, Rad, Rem.*) *(2)* A term used to express amount of exposure to a chemical substance.

Dose-Response Relationship: Correlation between the amount of exposure to an agent or toxic chemical and the resulting effect on the body.

Dosimeter (Dose Meter): An instrument used to determine the full-shift exposure a person has received to a physical hazard.

Dross: The scum forming on the surface of molten metals, largely oxides and impurities, such as the oxide scum that forms in a wavesolder machine.

Dry Etching: Controlled removal of a particular solid layer by use of a gas generated in a radio frequency field.

Duct Velocity: Air velocity through the duct cross section. When solid particulate material is present in the air stream, the duct velocity must exceed the minimum transport velocity.

E-Beam (Electron Beam) Evaporation: A process where heat from an electron beam is used to change a solid metal or alloy into a vapor that is deposited onto the wafer.

E-Beam (Electron Beam) Exposure: An exposure system used in mask making and lithography where a beam of electrons is used to directly generate the pattern without the use of a mask or reticle.

Electric Field: The term *electric field* is used in this book for the electric field strength. It is measured in volts per meter (V/m) as the true root-mean-square (RMS) value of the amplitude of the electric field strength at the measuring probe within a given measuring bandwidth and measuring time.

ELF: Extremely (or Extra) Low Frequency; the frequency range that is considered ELF varies. For example: the American Industrial Hygiene Association (AIHA) classifies frequencies of 30 Hz to 300 Hz as ELF, while the Institute of Electrical and Electronic Engineers (IEEE) considers the ELF range to be 5 Hz to 2 kHz.

Engineering Controls: A secondary control measure in which a hazard is mitigated by isolating the worker from the hazard (e.g., exhaust ventilation, glove boxes, ...).

Epitaxial Deposition: A CVD process where the deposited layer is the same crystalline structure as the substrate layer.

Ergonomic Hazards: Refers to workplace conditions that pose a biomechanical stress to the worker. Such hazardous conditions include, but are not limited to, faulty work station layout, improper work methods, improper tools, excessive tool vibration, and job design problems that include aspects of work flow, line speed, posture, and force required, work/rest regimens, and repetition rate.

Ergonomic Risk Factors: Conditions of a job, process or operation that contribute to the risk of developing CTDs. Examples include repetitiveness of activity, force required, and awkwardness of posture. Risk factors are regarded as synergistic elements of ergonomic hazards which must be considered in light of their combined effect in inducing CTDs.

Etching: Controlled removal of a particular solid layer by wet, dry chemical, or physical removal.

Etiologic Agent: Refers to organisms, substances, or objects associated with the cause of disease or injury.

Evaporation: A process where heat from a filament or electron beam is used to change a solid material such as a metal or alloy into a vapor that is deposited onto a wafer.

Exposure: A step within the lithographic process where ultraviolet light or another energy form is directed onto a resist-coated wafer to form a pattern.

Fabrication: Processes used to manufacture semiconductor devices.

Face Velocity: Average air velocity into the exhaust system measured at the opening into the hood or booth.

Fan Static Pressure: The pressure added to the system by the fan. It equals the sum of pressure losses in the system minus the velocity pressure in the air at the fan inlet.

Far Field (Free Field): In noise measurement, this refers to the distance from the noise source where the sound pressure level decreases 6 dBA for each doubling of distance (inverse square law).

Film Badge: A piece of masked photographic film worn by persons working with or near significant sources of ionizing radiation such as in a radiation-controlled area for a krypton^{85} fine leak detection unit. It is darkened by ionizing radiation, and exposure can be determined by inspecting the film.

Finishing Materials: Materials added to a new building near its completion (e.g., carpeting, partitions, ...).

Fit Test: A means for determining—either qualitatively or quantitatively—an individual's ability to obtain a good face fit with a particular respirator.

Flammable Liquid: Any liquid having a flash point below 100°F (37.8°C).

Flat: The part of an otherwise round wafer that has been ground down to allow proper orientation of the wafer during various process steps.

Flux: Usually refers to a substance used to clean surfaces and promote fusion in soldering.

Fume: Airborne particulate formed by the evaporation of solid materials, e.g., metal fume emitted during welding. usually less than one micron in diameter.

Gallium Arsenide (GaAs): The most common of the III-V wafer materials. Has the advantage of producing faster devices than the more common silicon wafers, but it is more expensive and harder to process.

Gamma Rays: A form of electromagnetic radiation emitted from radioactive substances. Many are highly penetrating, an appreciable fraction being able to penetrate several centimeters of lead.

Gauss (G): A Centimeter Gram Second unit for expressing magnetic flux density; 1 G = 1000 milliGauss; 1 G = 1 x 10^{-4} Teslas; 1 G = 0.1 milliTeslas. The Gauss is often used in the popular press and in medical literature. (Tesla is normally used in health physics literature.)

Geiger Counter: A gas-filled electrical device which counts the presence of an atomic particle or ray by detecting the ions produced. (Sometimes called a *Geiger-Mueller counter.*) For a portable instrument, it has good sensitivity.

General Ventilation: System of ventilation consisting of either natural or mechanically induced fresh air movements to mix with and dilute contaminants in the workroom air. This is not the recommended type of ventilation to control contaminants that are toxic.

Glycol Ethers and their Acetates: Made from the reaction of an oxide with an alcohol. Five glycol ethers have been targeted by the Semiconductor Industry Association for elimination because of their reproductive toxicity.

Glove Box: A sealed enclosure in which all handling of items inside the box is carried out through long impervious gloves sealed to ports in the walls of the enclosure.

Grab Sample: A sample which is taken within a very short time period. The sample is taken to determine the constituents at a specific time.

Hard Bake: A part of the lithography process where the exposed wafer is heated to evaporate off the carrier solvent remaining after the soft bake step. Temperatures used are higher than those used in the soft bake step increasing the adhesion of the resist at edges.

Hearing Conservation: The prevention or minimizing of noise-induced deafness through the use of hearing protection devices, the control of noise through engineering methods, annual audiometric tests, and employee training.

Hearing Level: The deviation in decibels of an individual's threshold from the zero reference of the audiometer.

HEPA (High Efficiency Particulate Air Filter): A filter with an accordion-like design that increases the surface area allowing a high filtration efficiency of at least 99.97 percent for 0.3 μ particles.

High Radiation Area: Any area, accessible to individuals, in which there exists ionizing radiation at such levels that an individual could receive, in any one hour, a dose in excess of 100 millirems to the whole body, head and trunk, active blood-forming organs, lens of eyes, or reproductive organs.

Homogeneous Exposure Group (HEG): A group of employees who experience chemical or biological agent exposures similar enough that monitoring agent exposure of any worker provides data useful for predicting exposures of the remaining workers.

Hood Static Pressure: The suction or static pressure in a duct near a hood. It represents the suction that is available to draw air into the hood.

HLF (Horizontal Laminar Flow) Hood: A fabrication or cleanroom work station that uses filtered laminar air from the side of the work station to keep particles off the work surface

Humidity: (1) Absolute humidity is the weight of water vapor per unit volume, pounds per cubic foot or grams per cubic centimeter; *(2)* relative humidity is the ratio of the actual partial vapor pressure of the water vapor in a space to the saturation pressure of pure water at the same temperature.

HVAC (Heating, Ventilating and Air Conditioning): The total air handling system for supplying and removing conditioned air to the work space.

IAQ (Indoor Air Quality): The term generally used to refer to chemical, physical and biological characteristics of indoor air in nonresidential environments, which can affect the comfort or health of the occupants.

IARC: International Agency for Research on Cancer

Immediately Dangerous to Life or Health (IDLH): Represents a maximum concentration from which, in the event of respirator failure, one could escape within 30 minutes without experiencing any escape-impairing or irreversible health effects.

Impingement: As used in air-sampling, impingement refers to a process for the collection of particulate matter in which a particle-containing gas is directed against a wetted glass plate and the particles are retained by the liquid.

Infrared Radiation: Electromagnetic energy with wavelengths from 770 nm to 12,000 nm.

Ingestion: The process of taking substances into the stomach, as food, drink, medicine, etc.

Integrated Circuit (IC): A circuit containing many interconnected elements on a individual semiconductor chip as opposed to a discrete device.

Inverse Square Law: The propagation of energy through space is inversely proportional to the square of the distance it must travel. An object 3 ft (m) away from an energy source receives 1/9 as much energy as an object 1 ft (m) away.

Ion Beam Milling: A dry etch process that is a physical etch procedure. An etch gas is ionized in a radio frequency field (usually 13.56 MHz), and then the individual molecules are accelerated to the wafer surface where material is removed by physical bombardment.

Ion Implantation: A doping process where ions are formed in a vacuum from a parent compound, accelerated to high speed, and bombarded into the wafer at precise locations.

Ionization: The process whereby one or more electrons is removed from a neutral atom by the action of radiation. Specific ionization is the number of ion pairs per unit distance in matter, usually air.

Ionization Chamber: A device roughly similar to a Geiger Counter and used to measure radioactivity.

Ionizing Radiation: Gamma rays and x-rays, alpha and beta particles, neutrons, protons, high speed electrons and other nuclear particles which will interact with gases, liquids, or solids to produce ions.

Isotropic: Exhibiting properties with the same values when measured along axes in all directions.

Isotropic Etching: Refers to any etching where the material is removed both downward and to the sides—as opposed to anisotropic etching. Often associated with wet etch processes.

Laminar Air Flow: Streamlined airflow in which the entire body of air within a designated space moves with uniform velocity in one direction along parallel flow lines.

Laser: Light Amplification by Stimulated Emission of Radiation; any device which can be made to produce or amplify electromagnetic radiation in the wavelength range greater than 200 nanometers (nm) but less than or equal to 1 mm primarily by the process of controlled stimulated emission.

Laser Safety Office (LSO): One who has the authority to monitor and enforce the control of laser hazards and effect the knowledgeable evaluation and control of laser hazards.

LC_{50} : Lethal airborne concentration that will kill 50 percent of the test animals within a specified time.

LD_{50} : The dose required to produce the death in 50 percent of the exposed species within a specified time.

Lithography: A critical semiconductor manufacturing process step where a pattern is formed on the wafer. When light is used to expose the resist material, the term used is photolithography. When x-rays are the exposure source, the term used is x-ray lithography. Microlithography is the term used when the pattern geometries are small enough to be measured in microns.

Local Exhaust Ventilation: A ventilation system that captures and removes the contaminants at the point they are being produced before they escape into the workroom air.

LSI: Large Scale Integration; integrated circuits with 5000 to 100,000 components.

Magnehelic®: An aneroid-type of guage suitable for the measurement of low pressures associated with local exhaust ventilation systems.

Magnetic Field (H field): The term magnetic field is used in this text for the magnetic flux density. It is measured in tesla (T) or gauss (G) as the root-mean-square (RMS) value of the magnetic flux density vector amplitude within given measuring bandwidths and measuring time.

Manometer: Instrument for measuring pressure; essentially a U-tube partially filled with a liquid (usually water, mercury, or a light oil), so constructed that the amount of displacement of the liquid indicates the pressure being exerted on the instrument.

Mask: A glass plate with the pattern that is transferred to the wafer during lithography.

Masking: The step in the lithography process where a precise pattern is transferred to the wafer. The pattern determines areas that are to be doped or selectively removed.

Maximum Concentration Values in the Workplace (MAKs): The maximum permissible concentration of a chemical compound present in the air within a working area (as gas, vapor, particulate matter) which, according to current knowledge, generally does not impair the health of the employee nor cause undue annoyance. Under these conditions, exposure can be repeated and of long duration over a daily period of eight hours, constituting an average work week of 40 hours (42 hours per week as averaged over four successive weeks for firms having four work shifts). Scientifically based criteria for health protection, rather than their technical or economical feasibility, are employed. These standards are set by the German Commission for the Investigation of Health Hazards of Chemical Compounds in the Work Area (Deutsche Forschungsgemeinschaft).

Maximum Exposure Limit (MEL): An airborne exposure limit for a substance for which the level of exposure should be reduced so far as is reasonably practicable and, in any event, should not exceed the MEL. These standards are set by the UK Health and Safety Executive.

Maximum Permitted Exposure (MPE): That level of laser radiation to which persons may be exposed without hazardous effect or adverse biological changes in the eyes or skin.

Membrane Filter: A filter medium made from various polymeric materials such as cellulose, polyethylene, or tetrapolyethylene. Membrane filters usually exhibit narrow ranges of effective pore diameters and, therefore, are very useful in collecting and sizing microscopic and submicroscopic particles and in sterilizing liquids.

Metallization: A process step where a layer of metal or alloy is deposited on the surface of the wafer to allow interconnection of components in a device. Aluminum is commonly used for this purpose on silicon wafers.

Monitoring Plan: A schedule of employee exposure monitoring which identifies what chemical or biological agents will be sampled, which jobs, tasks and/or areas will be sampled, the number of samples and frequency of monitoring, and nature of the samples (i.e., personal vs. area, full-shift TWA vs. STEL, etc.)

MSI: Medium Scale Integration; integrated circuits with 50 to 5000 components.

n-Type (Negative Type): A semiconductor material that contains more conduction electrons than holes making it positive. n-Type dopants for silicon are Group V elements (e.g., antimony, arsenic, phosphorous). n-Type dopants are generally more toxic than p-type dopants (e.g., boron).

National Institute for Occupational Safety and Health (NIOSH): U.S. government agency established to conduct research on the health effects of exposures in the work environment, to develop criteria for dealing with toxic materials and harmful agents, including "safe" levels of exposure, to train professional personnel for carrying out various responsibilities prescribed by the Occupational Safety and Health Act of 1970, and, in general, to conduct research and assistance programs for protecting and maintaining worker health.

NCRP: National Committee on Radiation Protection; an advisory group of scientists and professional people which makes recommendations for radiation protection in the U.S.

Near Field: In noise measurement, this refers to a field in the immediate vicinity of the noise source where the sound pressure level does not follow the inverse square law.

Negative Resist: A resist that hardens when exposed to an energy source such as UV light.

NIOSH Approved Method: Any method of sampling and analysis published in the National Institute for Occupational Safety and Health (NIOSH) Manual of Analytical Methods.

Occupational Exposure Standard (OES): An airborne exposure limit for a substance for which it is sufficient to ensure the level of exposure is reduced to that standard. These standards are set by the UK Health and Safety Executive.

Optical Density: A logarithmic expression for the optical attenuation produced by an attenuating material. Often used in the classification of laser protective eye wear.

p-Type (Positive Type): A semiconductor material that contains more holes than conduction electrons making it positive. p-Type dopants for silicon are Group III-A elements (e.g., boron).

Patterning: See *Masking.*

Permissible Exposure Limit (PEL): An airborne exposure limit for a substance that is the maximum permitted under U.S. Occupational Safety and Health Administration (OSHA) regulations.

Photohelic®: An electronic aneroid-type of guage which measures and records static pressures, as well as integrates velocity pressure directly to velocity, in a local exhaust ventilation system.

Photomasking: See *Masking.*

Photoresist: A resist that reacts to UV light.

Pitot Tube: A device consisting of two concentric tubes, one serving to measure the total or impact pressure existing in the airstream, the other to measure the static pressure only.

Plasma Etching: A dry chemical etch process where an etch gas is ionized in a radio frequency field (usually 13.56 MHz), and a then the individual molecules are accelerated to the wafer surface where material is removed by the chemical reaction of the ions in contact with the material to be etched.

Plasma: (1) The fluid part of the blood in which the blood cells are suspended. Also protoplasm. *(2)* A high-energy gas composed of ionized particles formed by introducing a parent gas or vapor into a high-energy radio frequency field.

Power Density: The intensity of electromagnetic radiation per unit area expressed as watts/cm^2.

PPE: Personal Protective Equipment; includes items such as gloves, goggles, respirators, and protective clothing that are used to protect the employee when other controls are not sufficient to reduce the risk.

Precision: The degree of agreement of repeated measurements of the same property, expressed in terms of dispersion of test results about the mean result obtained by repetitive testing of a homogeneous sample under specified conditions. The precision of a method is expressed quantitatively as the standard deviation computed from the results of a series of controlled determinations.

Professional Judgment: That capacity of an experienced professional to draw correct inferences from incomplete quantitative data, frequently on the basis of undefined subjective observations, analogy, and intuition.

Push-Pull Hood: A hood consisting of an air supply system on one side of the contaminant source blowing across the source and into an exhaust hood on the other side.

Pyrolysis: The breaking apart of complex molecules into simpler units by the use of heat, as in the pyrolysis of heavy oil into gasoline.

Qualitative Exposure Assessment: An exposure assessment based on integration of information and judgment, and not based on a rigorous quantitative analysis of data.

Quantitative Exposure Assessment: An exposure assessment based on rigorous analysis of actual, quantitative measurements of exposure.

Radiation Area: Based on U.S. regulations: any area, accessible to individuals, in which there exists ionizing radiation at such levels that an individual could receive in any one hour a dose to the whole body in excess of 5 millirems, or in any 5 consecutive days a dose in excess of 100 millirems.

Radiation Machine: Any device capable of producing ionizing radiation when the associated control devices are operated, but excluding devices which produce radiation only by the use of radioactive material. Most electrical equipment operating at or above 15 kV is considered a radiation machine (e.g., electron microscopes, ion implanters...).

Radioactive Material: Any material which emits ionizing radiation spontaneously.

Reactive Ion Etching (RIE): A dry etch process that is a combination of a dry chemical and physical etch procedure. An etch gas is ionized in a radio frequency field (usually 13.56 MHz), and then the individual molecules are accelerated to the wafer surface where material is removed by both chemical reaction and physical bombardment.

Recommended Exposure Limit (REL): A recommended airborne exposure limit for a hazardous substance developed by the U.S. National Institute for Occupational Safety and Health. Often RELs are time-weighted average (TWA) concentrations for up to a 10-hour workday during a 40-hour workweek.

Relative Humidity: The ratio of the quantity of water vapor present in the air to the quantity that would saturate it at any specific temperature.

Rem (Roentgen Equivalent Man): A unit of dose of any ionizing radiation to body tissue in terms of its estimated biological effects relative to an exposure of one roentgen of x-rays or gamma rays.

Any of the following is considered to be equivalent to a dose of one rem:

(1) An exposure of 1 roentgen due to x- or gamma radiation.

(2) A dose of 1 rad due to x-ray, gamma or beta radiation.

(3) A dose of 0.1 rad due to neutrons or high energy protons.

(4) A dose of 0.05 rad due to particles heavier than protons and with sufficient energy to reach the lens of the eye.

One roentgen equals 2.58 x 10^{-4} coulomb per kilogram of air.

Representative: The degree to which a sample is, or samples are, characteristic of the whole medium exposure, or dose for which the samples are being used to make inferences.

Resist: A energy-sensitive film spun on to the wafer and exposed to an energy source (e.g., UV light, x-rays, or electrons) through a mask during the lithography process. The exposed resist (in the case of positive resists) or unexposed resist (in the case of negative resist) is dissolved with developers leaving a pattern of resist which allows etching to occur in some areas.

Respirator: Any device designed to provide the wearer with respiratory protection against inhalation of a hazardous atmosphere. This includes, but is not limited to, air-supplied breathing apparatus, cartridge air-purifying respirators, maintenance-free vapor masks, and disposable dust masks with an assigned protection factor.

Reticle: (1) A reproduction of the pattern that is to be imaged on a wafer (a lithography step) or imaged on a mask (a mask making step). *(2)* A scale or grid or other pattern located in the focus of the eyepiece of the microscope.

Rinse: The removal of an active chemical such as a wet etchant or developer from the surface of the wafer by any of several means such as immersion in a DI water bubbler or spray rinsing.

SBS: Sick Building Syndrome; a term used to describe nonspecific symptoms; of discomfort and ill health, such as eye, nose and throat irritation, headaches, unspecified allergies, nausea, fatigue, skin rashes and other similar complaints. The symptoms are general and subjective and may be indicative of other medical conditions.

Sealed Source: Any radioactive material that is permanently encapsulated in such a manner that the radioactive material will not be released under the most severe conditions likely to be encountered by the source.

Service: Performance of adjustments, repairs or procedures required to return the unit to its intended state. Service may affect the classification of the equipment.

Silicon (Si): The most common base semiconductor wafer material. A nonmetallic element which, after oxygen, is the main elementary constituent of the earth's crust.

Soft Bake: A part of the lithography process where the exposed wafer is heated to evaporate some of the carrier solvent in the resist leaving the resist soft.

Solder: A material used for joining metal surfaces together by filling a joint or covering a junction. The most commonly used solder is one containing lead (37%) and tin (63%). Silver solder may contain cadmium as a contaminant. Zinc chloride and fluorides are commonly used as fluxes to clean the surfaces.

Sound Level: A weighted sound pressure level, obtained by the use of metering characteristics and the weighting A, B, or C specified in ANSI S1.4.

Sputtering: A method of depositing a thin layer (typically metal) on a substrate by bombarding a target material with radio-frequency-generated ions. The frequency typically used is 13.56 MHz.

SSI: Small Scale Integration; integrated circuits with between 2 and 50 components.

Standard Conditions: In industrial ventilation, 21.1°C (70°F), 50 percent relative humidity, and 101.3 kPa (29.92 in. of mercury) atmospheric pressure.

Stepper: A machine used in the lithography process to transfer a pattern from a reticle on to a resist-covered wafer in a series of individual exposure steps.

Stripping: In semiconductor manufacturing, refers to the removal of a resist layer by wet chemicals, such as an organic acid mixture, or a dry plasma system.

Substitution: A primary control measure in which a hazardous material or operation is replaced by a less hazardous material or operation.

Substrate: Underlying surface of a wafer on which a layer is formed.

TBS: Tight Building Syndrome; see *SBS.*

Temporary Threshold Shift (TTS): The hearing loss suffered as the result of noise exposure, all or part of which is recovered during an arbitrary period of time when one is removed from the noise. It accounts for the necessity of checking hearing acuity at least 16 hours after a noise exposure.

Teratogen: An agent or substance that may cause physical defects in the developing embryo or fetus when a pregnant female is exposed to that substance.

Threshold Limit Value (TLV): Refers to airborne concentrations of substances or levels of physical agents and represents conditions under which it is believed that nearly all workers may be exposed day after day without adverse effect. These standards are set by the ACGIH.

Threshold Limit Value-Time Weighted Average (TLV-TWA): Time-weighted average concentration for a normal 8-hour workday and a 40-hour workweek.

Threshold Limit Value-Short Term Exposure Limit (TLV-STEL): The concentration to which workers can be exposed continuously for a short period of time without adverse health effects. It is a 15-minute time-weighted average exposure. Exposures at the STEL should not be longer than 15 minutes and should not be repeated more than four times per day with at least 60 minutes between successive exposures at the STEL.

Threshold Limit Value-Ceiling (TLV-C): The concentration that should not be exceeded during any part of the working exposure.

Time-Weighted Average Concentration (TWA): Refers to concentrations of airborne toxic materials which have been weighted for a certain time duration, usually 8 hours.

ULSI: Ultra Large Scale Integration; integrated circuits with greater than 1 million components.

Velocity Pressure: The kinetic pressure in the direction of flow necessary to cause a fluid at rest to flow at a given velocity. When added to static pressure, it gives total pressure.

Velometer: A device for measuring air velocity.

Visible Radiation: The wavelengths of the electromagnetic spectrum between 400–700 nm.

VLF (Vertical Laminar Flow) Hood: A fabrication or cleanroom work station that uses filtered laminar air from the top of the work station to keep particles off the work surface.

VLF: Very Low Frequency; the frequency range that is considered VLF varies. For example: the AIHA classifies frequencies of 3 kHz to 30 kHz as VLF, while the IEEE considers the VLF range to be 2 kHz to 400 kHz.

VOC: Volatile Organic Compounds; a ubiquitous class of indoor pollutants which originate from pressed wood products, insulation, adhesives, textiles, and paints. Examples are formaldehyde, and chlorinated and aromatic hydrocarbons.

Wafer: A thin, typically round (except for the flat) semiconductor base material.

Wet Etching: Controlled removal of a particular solid layer by immersion in a liquid such as buffered oxide etch. The liquid may be heated or kept a room temperature.

Wipe Samples: Samples taken by wiping a specific area (often 100 cm^2) with moistened filter paper to semi-quantitatively characterize surface contamination.

X-ray Exposure: (1) An exposure system used in lithography where x-rays are directed through a mask or reticle on to a resist to form a pattern. *(2)* An employee exposure to x-rays.

X-ray: A type of electromagnetic radiation of higher frequency than visible light. Usually produced by high-speed electrons impinging on a metal target. x-rays are similar to gamma rays, but do not come from the nucleus of the atom but from the surrounding electrons.

Yield: The amount of usable product as a percentage of the product that started the process step.